

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions

Remote

- Built a resilient **Puppeteer/Node.js workflow engine** for MCA company-registration filings, reducing manual filing time by **85%** through state recovery, retries, and public-site API lookups.
- Optimized an **AI-assisted company-registration pipeline** for unique company-name suggestions, descriptions, and filing-field drafts, reducing manual drafting/rework across MCA workflow preparation.
- Containerized services with **Docker** and automated **GitHub Actions** deployments to **Google Cloud Run**, improving release velocity by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built MRI segmentation tooling for **95 anatomical regions**, turning model outputs into usable masks and 3D visualization artifacts inside an existing application workflow.
- Reduced processing runtime by **87%** with async/parallel execution and fewer repeated post-processing passes over large MRI volumes.
- Integrated outputs into 3D reconstructed surface visualization flows so model results stayed inspectable and useful to end users.

Open Source

[Contributions](#) 

- Contributed to Invesalious, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, working with existing maintainers, code style, and review constraints.

Projects

Murdoc | *Python, FastAPI, MCP, OpenTelemetry, Prometheus, Docker*

[GitHub](#) 

- Built a gateway that preserves context around prompts, tools, retrieved data, policy decisions, and outputs, making AI-agent actions inspectable through structured traces and audit ledgers.
- Modeled runtime settings, route profiles, RBAC decisions, and attack-lab outcomes as explicit metadata so agent behavior could be governed, replayed, and debugged.

Penny Lane Capital (Tradeage) | *Python, Agno, Pydantic, SQLite, yfinance, JSONL*

- Built an **Agno Workflow** multi-agent research system with analyst roles, bull/bear debate, trader, risk review, and portfolio approval using Pydantic state, checkpoints, memory, and JSONL traces.
- Evaluated agent decisions with historical-market backtests, reward loops, validation harnesses, and **5 baselines**, producing return, drawdown, and win-rate scorecards across market windows.

Swish Support System | *Python, Go, FastAPI, Gin, PostgreSQL, Redis, Langfuse*

[GitHub](#) 

- Built AI-assisted backend workflows for order status, delays, refunds, trust checks, and escalation using tool calls over operational APIs with PostgreSQL and Redis state.
- Added retrieval-backed responses, confidence thresholds, human handoff paths, conversation memory, and traceable action records for ambiguous or high-risk user workflows.

Technical Skills

Languages: Python, Go, TypeScript

Frameworks: FastAPI, Gin, Echo, Node.js, Agno

Tools: PostgreSQL, Redis, RabbitMQ, Docker, Langfuse, OpenTelemetry, GitHub Actions, AWS